Euler's number

1 Consider the function $f(x) = e^x$.

a Complete the following table of values, correct to two decimal places:

x	-3	-2	-1	0	1	2	3
$f(x)$							

b Sketch the graph of $f(x) = e^x$.

2 Sketch the curves $y = e^x$, $y = e^x + 3$, and $y = e^x - 4$ on the same coordinate plane.

3 Use a calculator or other technology to approximate the each of the following values correct to four decimal places:

a e^4 **b** e^{-1} **c** $e^{\frac{1}{5}}$ **d** $5\sqrt{e}$

e $\dfrac{4}{e}$ **f** $\dfrac{8}{9e^4}$

The natural logarithm

4 Find the value of each of the following correct to four decimal places:

a $\ln 94$ **b** $\ln 0.042$ **c** $\ln 78^4$ **d** $\ln(18 \times 35)$

5 Consider $x = \ln 31$. Find the value of x, correct to two decimal places.

6 Find the value of $5 \log_e e$.

7 Is the value of $\log_e 2$ greater than or less than 1?

8 Evaluate each of the following expressions:

a $\ln e^{3.5}$ **b** $\ln e^4$ **c** $\sqrt{6} \ln\left(e^{\sqrt{6}}\right)$ **d** $\ln\left(\dfrac{1}{e^2}\right)$

9 Write the following equations in logarithmic form.

a $e^0 = 1$ **b** $e^x = 3$ **c** $e^{\frac{1}{2}} = \sqrt{e}$ **d** $e^p = q$

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Applications

10 The population P of a city increases according to the formula $P = 3000e^{kt}$ where t is measured in years and k is a constant.

a Find the initial population.

b Given the population increases to 8000 in 2 years find the exact value of k.

c How many complete years will it take for the population to at least double?

11 The voltage V in volts across an electrical component decays according to the equation $V = Ae^{-\frac{1}{2}t}$ where A is the initial voltage and t is the time in years. Find the value of t such that V is half of the initial voltage, correct to two decimal places.

12 The spread of a virus through a city is modelled by the function: $N = \dfrac{15\,000}{1 + 100e^{-0.5t}}$, where N is the number of people infected by the virus after t days.

a How many people will have been infected after 3 days? Round your answer to the nearest whole number.

b How many whole days will it take for at least 4000 people to be infected with the virus?

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